

WEEK 4 ASSIGNMENT

Azure Cloud & Data Pipeline using Azure Data Factory

Name:
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Internship:
Celebal Technologies

Week:
Week 4

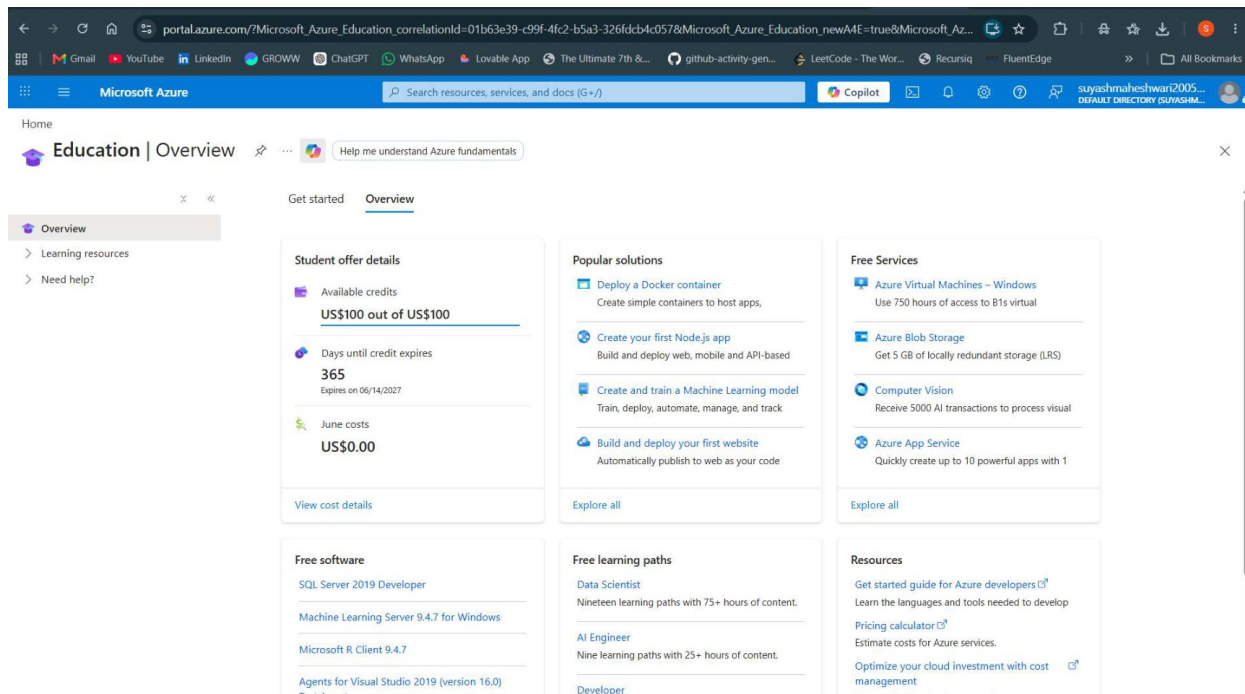
Date:
June 15, 2026

Task 1: Azure Portal & Resource Group

Logged into Azure for Students (portal.azure.com) with \$100 free credits. Created resource group rg-week4-pipeline in the Azure for Students subscription.

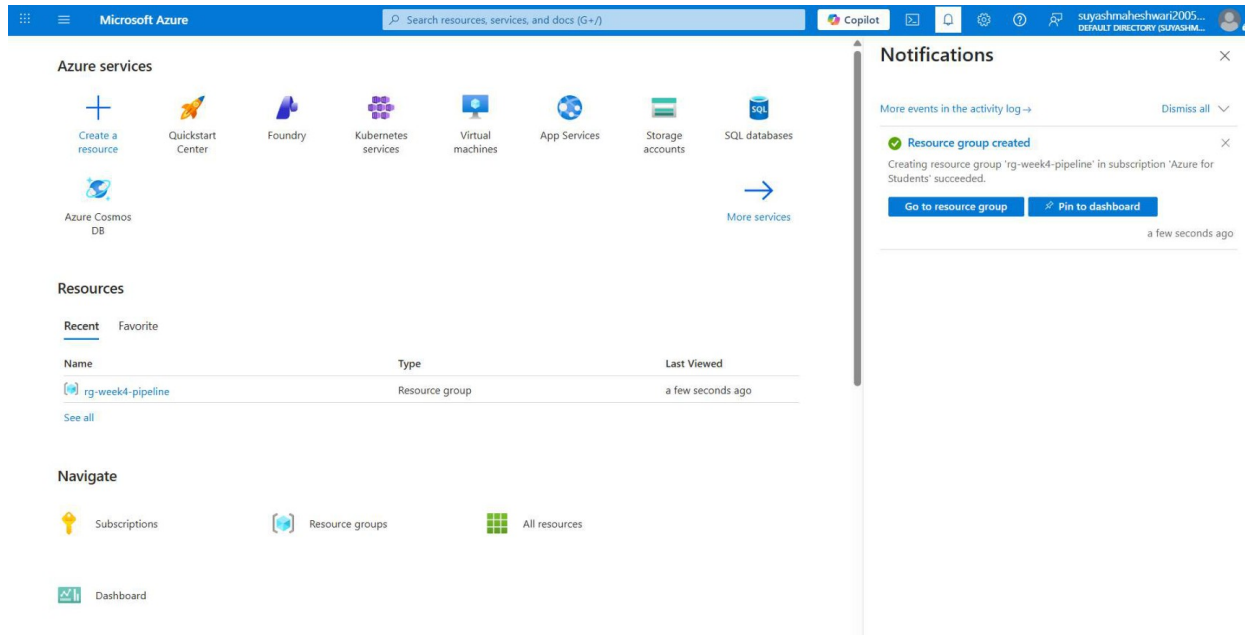
Screenshot 1.1 – Azure for Students Dashboard

 Azure Education Overview – US\$100 credits, 365 days validity, June costs: \$0.00



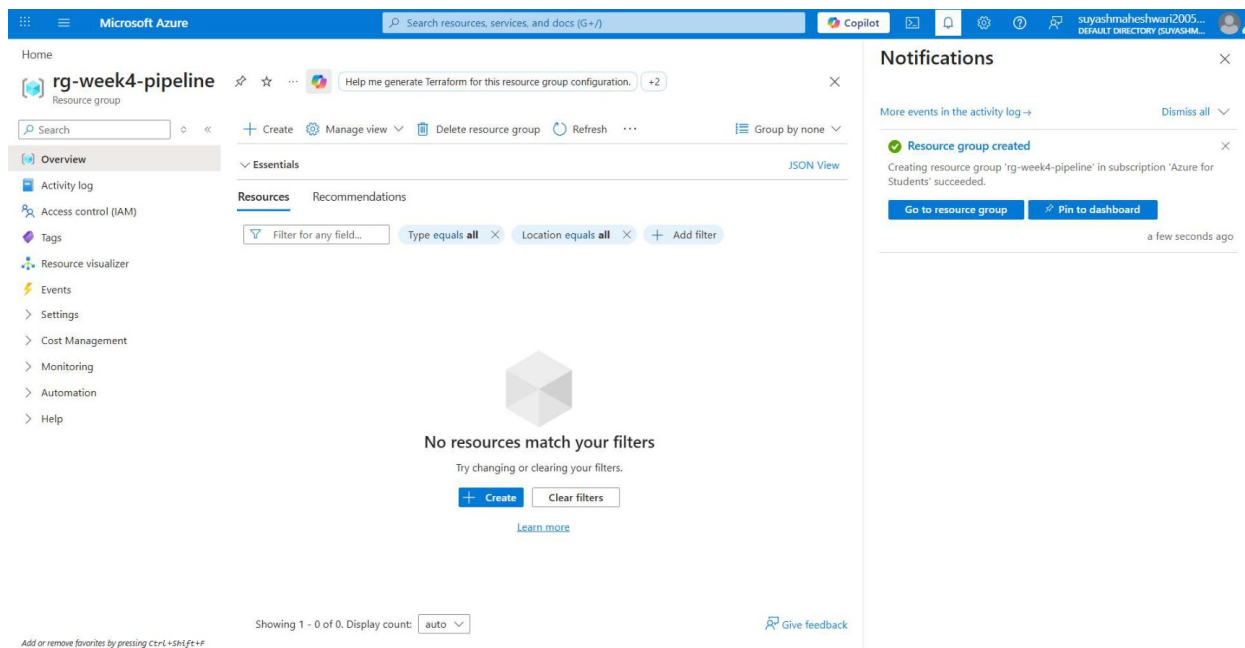
Screenshot 1.2 – Resource Group Created (Notification)

 Notification: Resource group 'rg-week4-pipeline' created successfully in Azure for Students



Screenshot 1.3 – Resource Group Overview

rg-week4-pipeline | Overview – Resource group successfully provisioned



Task 2: Storage Account, Blob Container & CSV Upload

Created Storage Account week4storageadf in Central India (East US restricted for Azure for Students). Created two containers: input-data (source) and output-data (destination). Uploaded Sample - Superstore.csv (2.18 MiB) to input-data.

Screenshot 2.1 – Storage Account Deployment Complete

📌 week4storageadf | Deployment complete – Resource group: rg-week4-pipeline, Subscription: Azure for Students

The screenshot shows the Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile 'suyashmaheshwari2005@...'. The main content area displays the 'Overview' page for a deployment named 'week4storageadf_1781460478674'. A green checkmark indicates 'Your deployment is complete'. The deployment details show the name, subscription ('Azure for Students'), resource group ('rg-week4-pipeline'), start time ('14/06/2026, 23:38:12'), and correlation ID. A 'Go to resource' button is visible. On the right, there are several informational cards: 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Home

week4storageadf_1781460478674 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : week4storageadf_1781460478674

Subscription : Azure for Students

Resource group : rg-week4-pipeline

Start time : 14/06/2026, 23:38:12

Correlation ID : 055cf600-d81b-4536-9f82-4efc747ef513

> Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. Find an Azure expert >

Screenshot 2.2 – Input Container with CSV File

📌 input-data container | Sample - Superstore.csv – 2.18 MiB, Block blob, Hot tier, Available

The screenshot shows the Azure portal interface for the 'input-data' container. The top navigation bar is the same as the previous screenshot. The main content area displays the 'Overview' page for the 'input-data' container. It shows the authentication method as 'Access key' and a search filter. A table lists the blobs in the container, showing one item: 'Sample - Superstore.csv'.

Home > week4storageadf | Containers

input-data

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

input-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

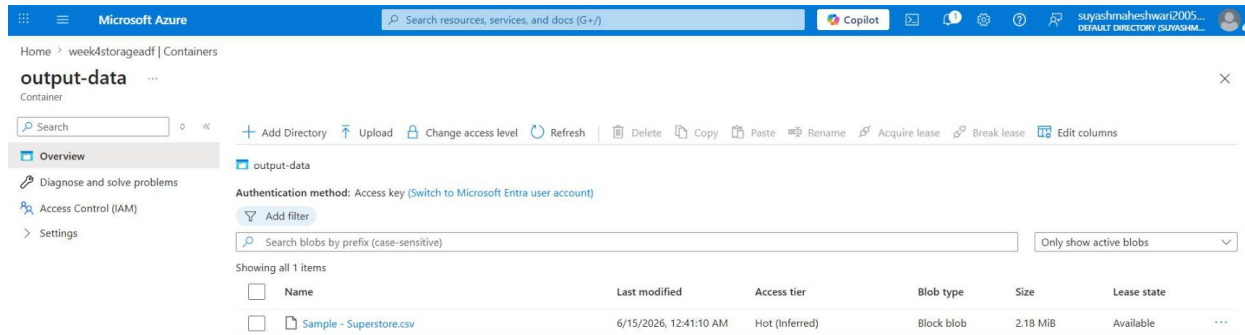
Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - Superstore.csv	6/14/2026, 11:54:17 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Screenshot 2.3 – Output Container (Destination)

📌 output-data container | Sample - Superstore.csv copied here by ADF pipeline (2.18 MiB, 6/15/2026 12:41 AM)

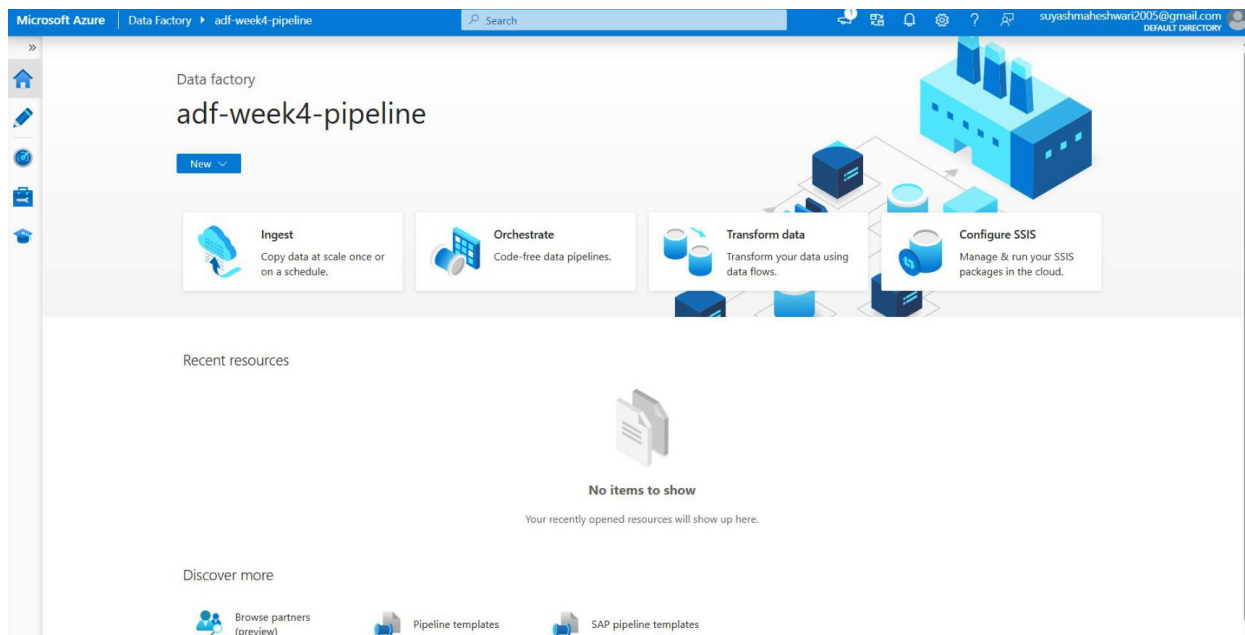


Task 3: ADF – UI Exploration, Linked Service & Datasets

Created Azure Data Factory adf-week4-pipeline. Launched ADF Studio and explored Author, Monitor, Manage tabs. Created Linked Service and source/destination datasets with Get Metadata activity.

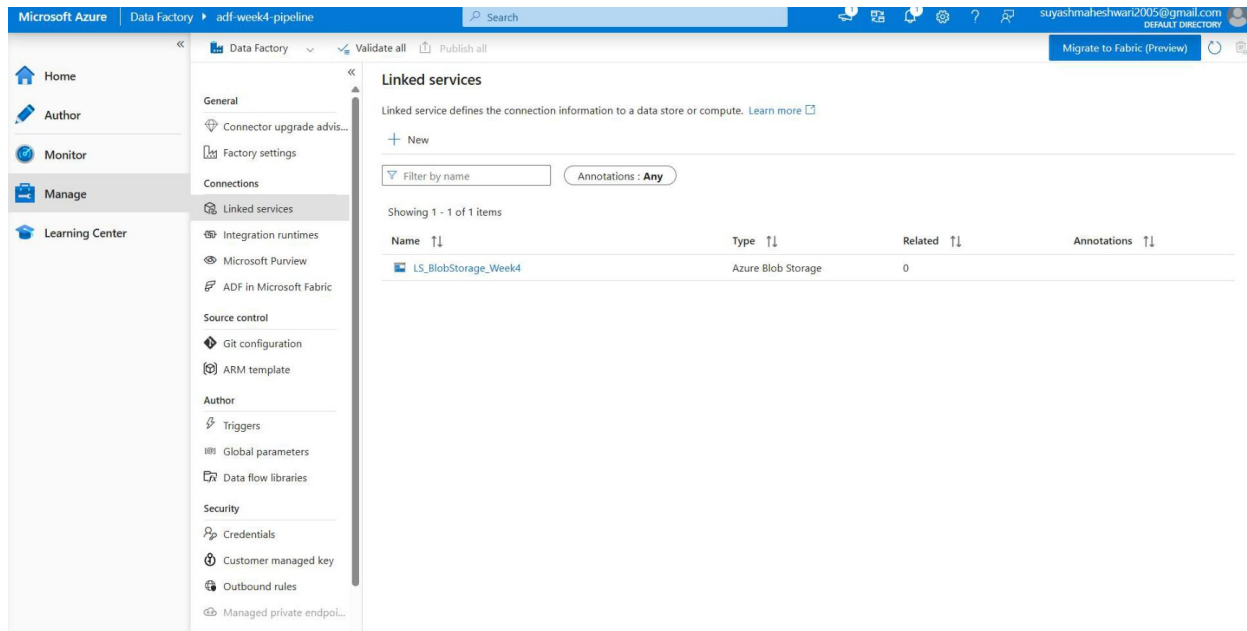
Screenshot 3.1 – ADF Studio Home (Author Tab)

adf-week4-pipeline | ADF Studio – Ingest, Orchestrate, Transform data, Configure SSIS options visible



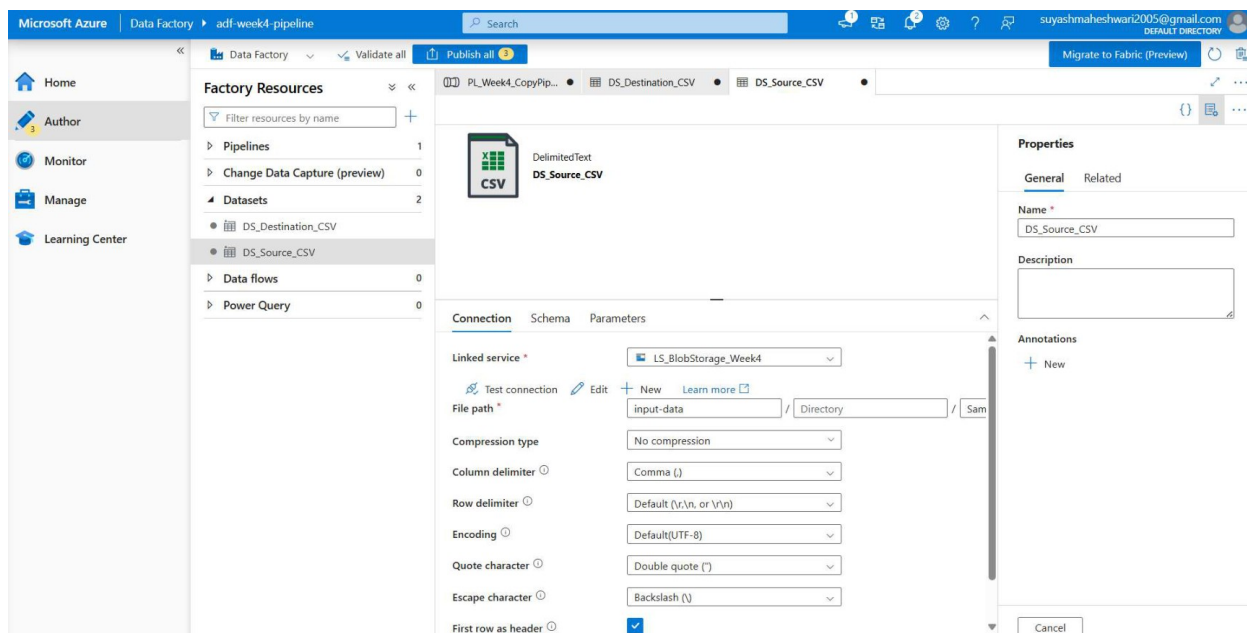
Screenshot 3.2 – Linked Service: LS_BlobStorage_Week4

Manage → Linked Services | LS_BlobStorage_Week4 (Azure Blob Storage) – Showing 1 of 1 items



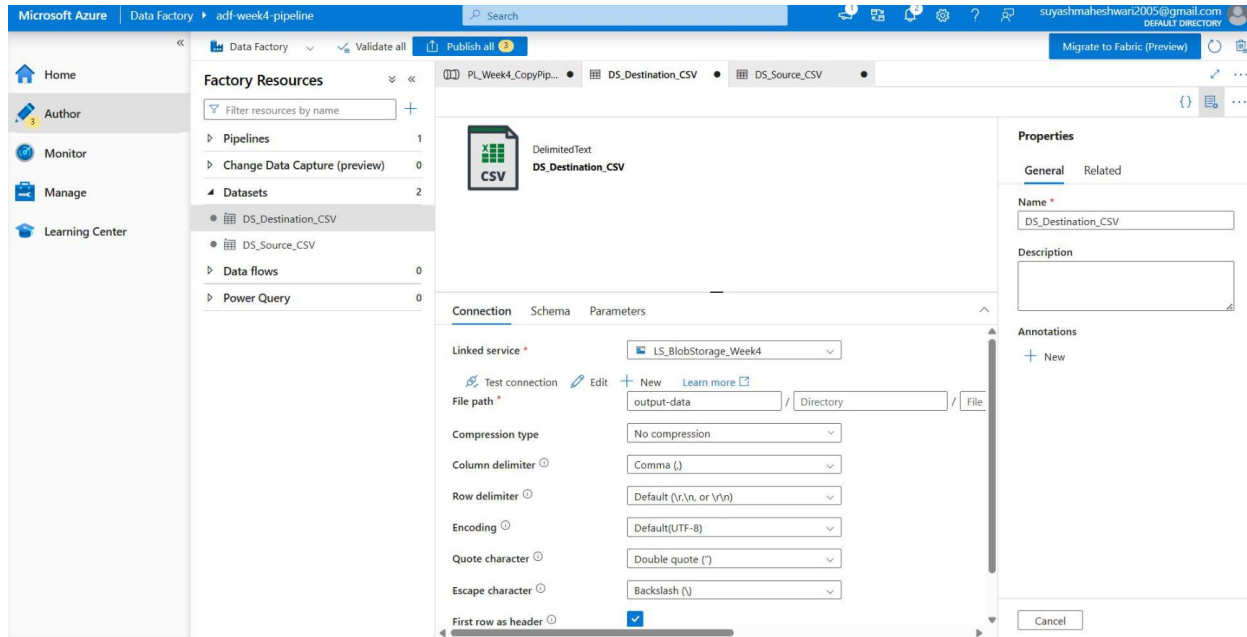
Screenshot 3.3 – Source Dataset: DS_Source_CSV

📌 DS_Source_CSV | DelimitedText – Linked service: LS_BlobStorage_Week4, File path: input-data / Sam...



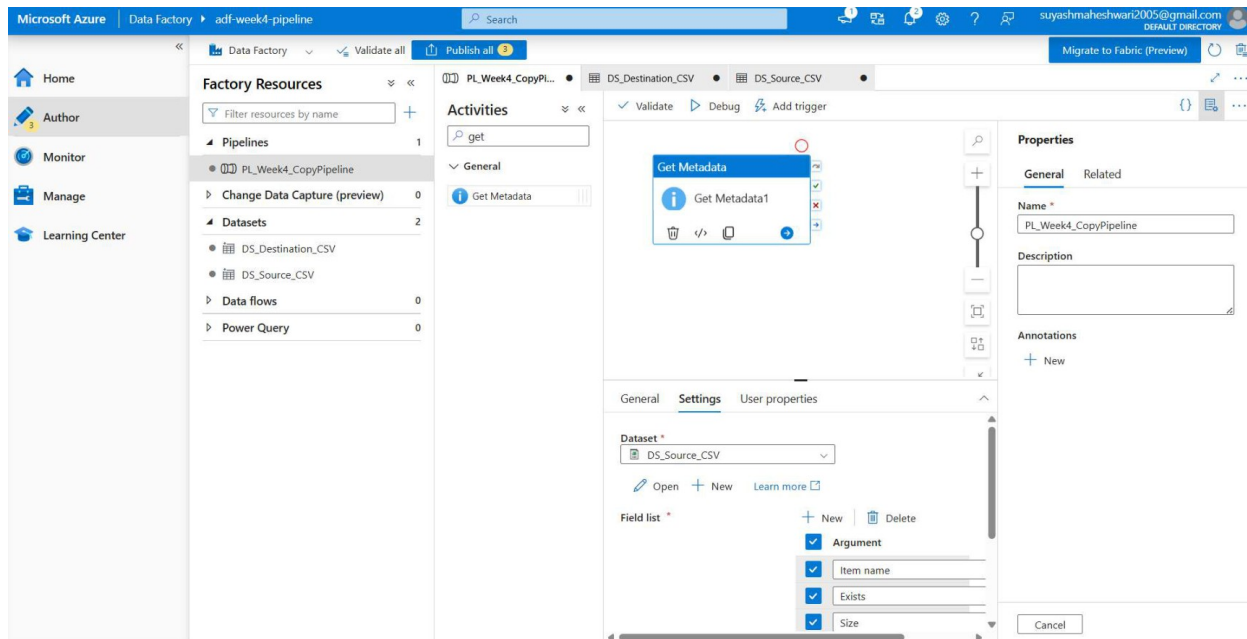
Screenshot 3.4 – Destination Dataset: DS_Destination_CSV

📌 DS_Destination_CSV | DelimitedText – Linked service: LS_BlobStorage_Week4, File path: output-data



Screenshot 3.5 – Get Metadata Activity Configured

📌 PL_Week4_CopyPipeline | Get Metadata1 – Dataset: DS_Source_CSV, Field list: Item name, Exists, Size

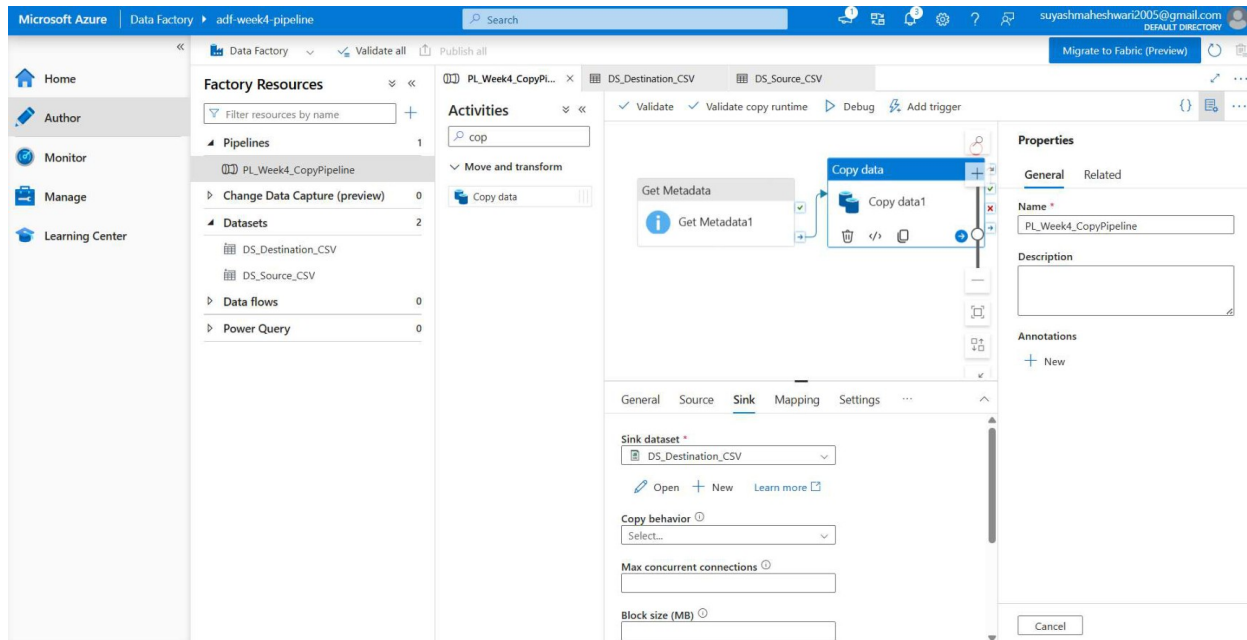


Task 4: Pipeline Design with Copy Data Activity

Built PL_Week4_CopyPipeline with Get Metadata → Copy Data connected via success dependency. Source: DS_Source_CSV. Sink: DS_Destination_CSV.

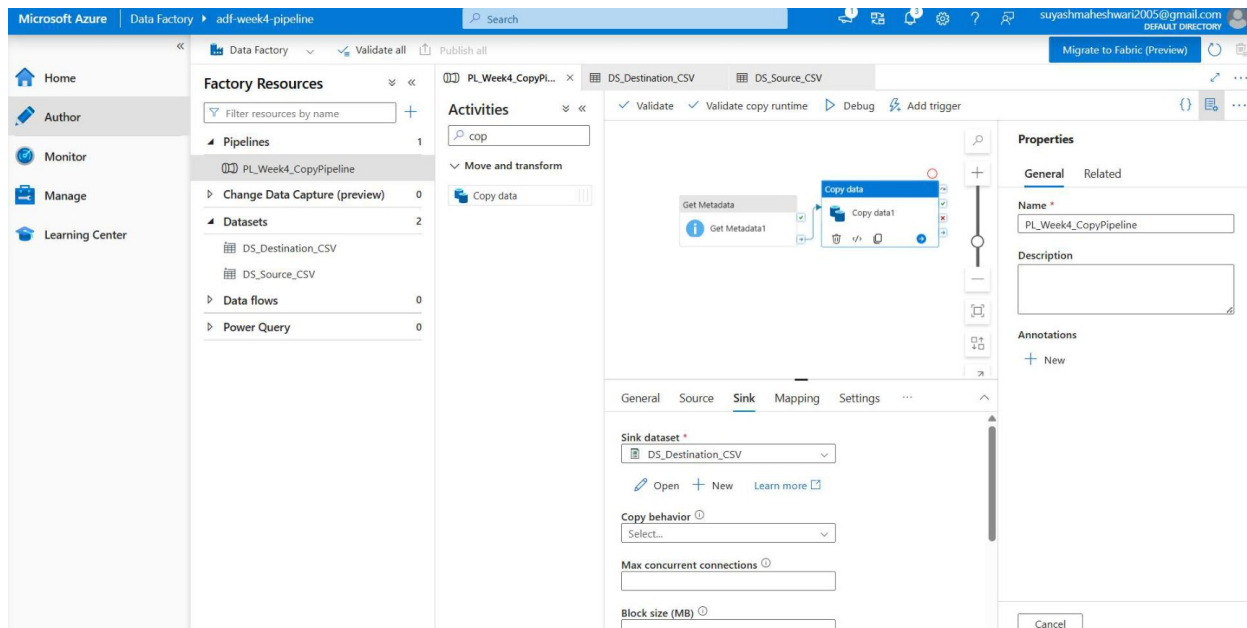
Screenshot 4.1 – Pipeline Canvas: Get Metadata → Copy Data

PL_Week4_CopyPipeline | Canvas – Get Metadata1 connected to Copy data1 with success arrow



Screenshot 4.2 – Copy Data Sink Configured

Copy data1 | Sink tab – Sink dataset: DS_Destination_CSV (output-data container)



Task 5: Pipeline Execution & Monitoring

Executed via Debug run (both activities Succeeded) then Trigger Now. Monitored in Monitor tab. Output file verified in output-data container.

Screenshot 5.1 – Debug Run: Pipeline Status Succeeded

🔴 PL_Week4_CopyPipeline | Debug Output – Pipeline status: ✔ Succeeded | Copy data1: Succeeded | Get Metadata1: Succeeded

The screenshot shows the Microsoft Azure Data Factory Author interface. The left sidebar contains navigation links: Home, Author, Monitor, Manage, and Learning Center. The main area displays the 'Factory Resources' on the left, including Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Activities' section shows a 'Move and transform' activity named 'Copy data'. The 'Properties' pane on the right shows the pipeline details: Name 'PL_Week4_CopyPipeline', Description, and Annotations. The 'Output' tab is selected, showing the pipeline status as 'Succeeded' and a table of activity results.

Activity name	Activity status	Activity name	Activity status
Copy data1	✔ Succeeded	Copy data	6/
Get Metadata1	✔ Succeeded	Get Metadata	6/

Screenshot 5.2 – Monitor: Pipeline Run Succeeded

🔴 Monitor → Pipeline Runs | PL_Week4_CopyPipeline – ✔ Succeeded | Duration: 32s | Triggered by: Manual trigger

The screenshot shows the Microsoft Azure Data Factory Monitor interface. The left sidebar contains navigation links: Home, Author, Monitor, Manage, and Learning Center. The main area displays the 'Pipeline runs' section. The 'Triggered' tab is selected, showing a table of pipeline runs. The table has columns for Pipeline name, Run start, Run end, Duration, Triggered by, Status, Run, and Parameters. The first row shows the pipeline 'PL_Week4_CopyPipeline' with a run start of 6/15/2026, 12:40:44 AM, a run end of 6/15/2026, 12:41:15 AM, a duration of 32s, triggered by 'Manual trigger', and a status of 'Succeeded'.

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters
PL_Week4_CopyPipeline	6/15/2026, 12:40:44 AM	6/15/2026, 12:41:15 AM	32s	Manual trigger	✔ Succeeded	Original	

Screenshot 5.3 – Activity Runs (Trigger Run)

🔴 PL_Week4_CopyPipeline Activity Runs | Get Metadata1: ✔ Succeeded (6s) | Copy data1: ✔ Succeeded (15s)

Microsoft Azure | Data Factory | adf-week4-pipeline

Search

Home | Author | Monitor | Manage | Learning Center

Dashboards | Runs | Pipeline runs | Trigger runs | Change Data Capture (previ...) | Runtimes & sessions | Integration runtimes | Data flow debug | Notifications | Alerts & metrics

All pipeline runs > PL_Week4.CopyPipeline - Activity runs

Rerun | Cancel | Refresh | Update pipeline | List | Gantt

Get Metadata | Copy data

Get Metadata1 | Copy data1

Activity runs

Pipeline run ID: 88a25432-1ed0-49b4-b447-4480dd09a595

All status | Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 items

Activity name	Activity st...	Activity type	Run start	Duration	Integration runtime	User pr
Copy data1	Succeeded	Copy data	6/15/2026, 12:40:59 AM	15s	AutoResolveIntegrationRuntime (Central India)	
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 12:40:52 AM	6s	AutoResolveIntegrationRuntime (Central India)	

Screenshot 5.4 – Monitor Activity Runs Detail View

Full activity runs view – Both activities shown Succeeded with green ticks on pipeline canvas

Microsoft Azure | Data Factory | adf-week4-pipeline

Search

Home | Author | Monitor | Manage | Learning Center

Dashboards | Runs | Pipeline runs | Trigger runs | Change Data Capture (previ...) | Runtimes & sessions | Integration runtimes | Data flow debug | Notifications | Alerts & metrics

All pipeline runs > PL_Week4.CopyPipeline - Activity runs

Rerun | Cancel | Refresh | Update pipeline | List | Gantt

Get Metadata | Copy data

Get Metadata1 | Copy data1

Activity runs

Pipeline run ID: 88a25432-1ed0-49b4-b447-4480dd09a595

All status | Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 items

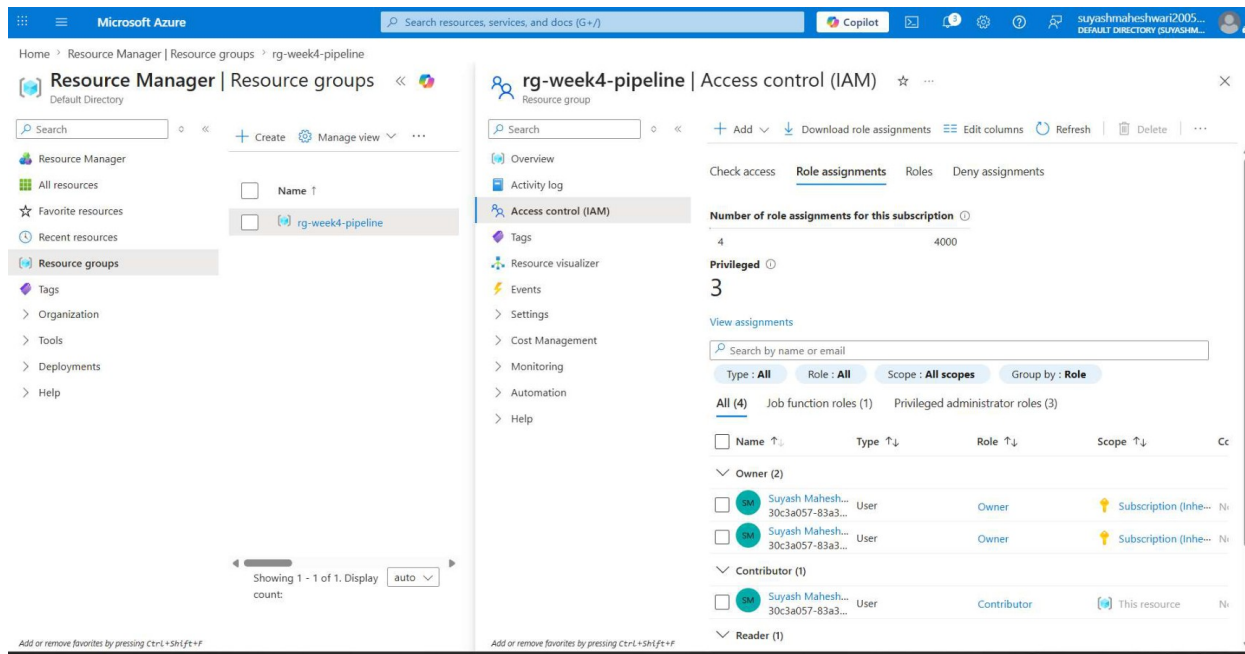
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User pr
Copy data1	Succeeded	Copy data	6/15/2026, 12:40:59 AM	15s	AutoResolveIntegrationRuntime (Central India)	
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 12:40:52 AM	6s	AutoResolveIntegrationRuntime (Central India)	

Task 6: IAM Role Assignment

Assigned IAM roles on rg-week4-pipeline: Owner (inherited), Contributor, Reader for user. Assigned Storage Blob Data Contributor to ADF Managed Identity for secure key-free access to Storage Account.

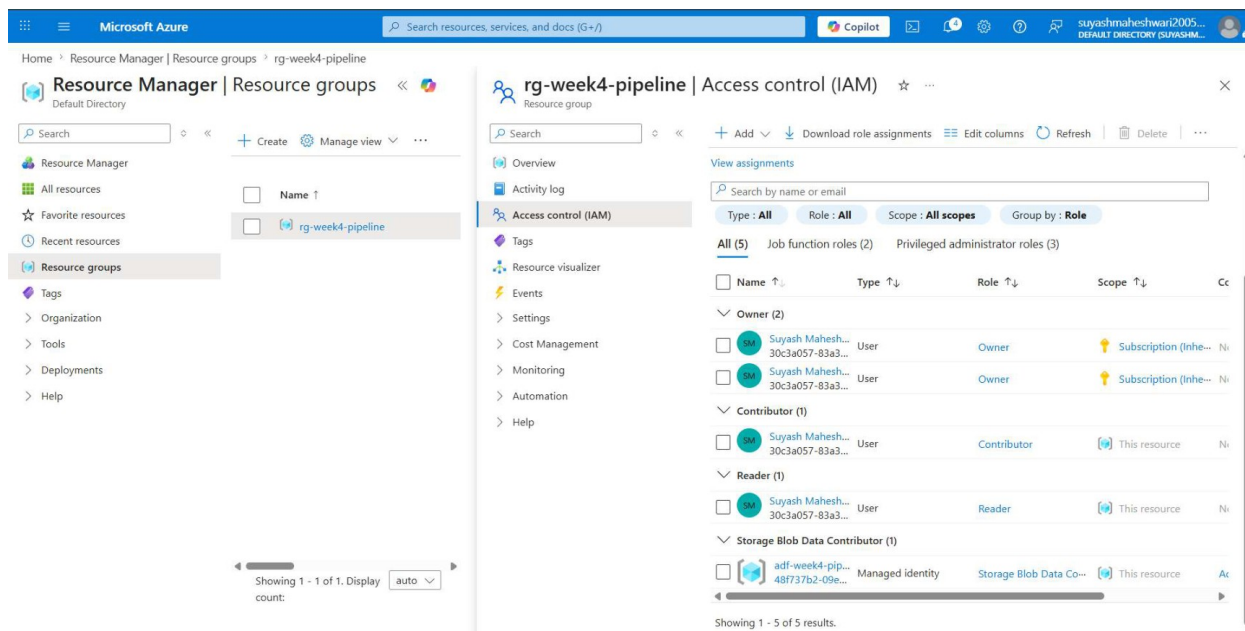
Screenshot 6.1 – IAM Role Assignments (Initial)

rg-week4-pipeline | Access Control (IAM) – Role assignments: Owner (2), Contributor (1), Reader (1) = 4 total



Screenshot 6.2 – All 5 IAM Roles Assigned

rg-week4-pipeline | IAM – All 5 roles: Owner, Contributor, Reader + Storage Blob Data Contributor for adf-week4-pipeline Managed Identity



Mini Project: End-to-End Data Pipeline

Problem Statement

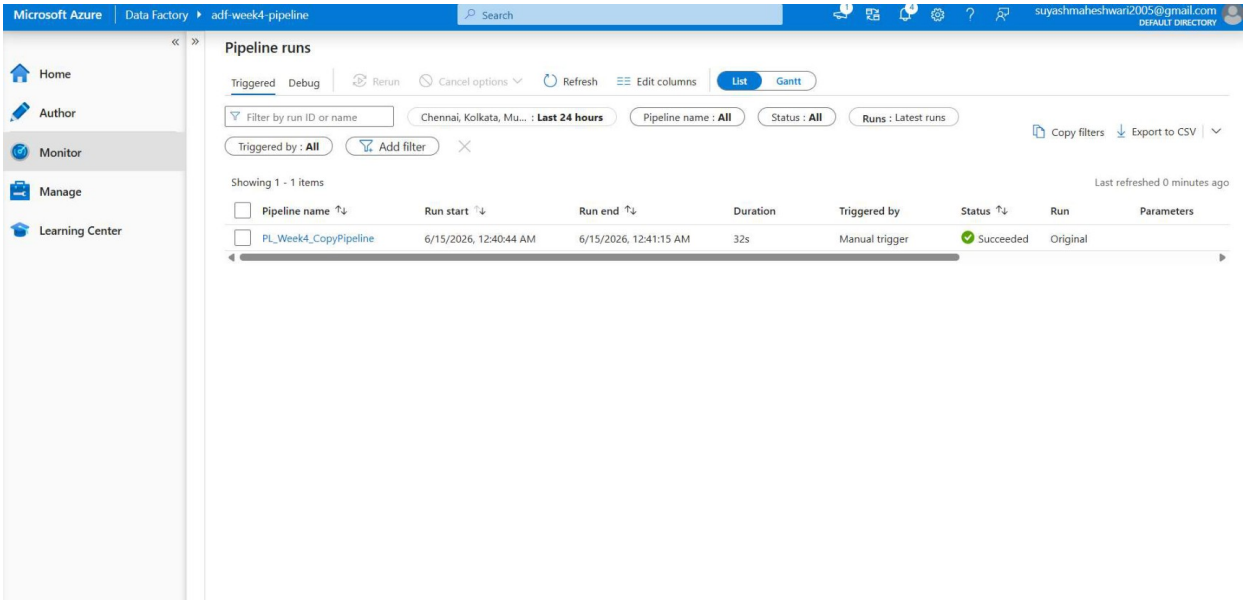
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory — with Linked Service, Dataset, Get Metadata validation, Copy Data, and destination output.

Component Summary

Component	Configuration	Status
Resource Group	rg-week4-pipeline (Azure for Students)	✔ Created
Storage Account	week4storageadf – Central India, Standard LRS	✔ Deployed
Source Container	input-data / Sample - Superstore.csv (2.18 MB)	✔ Uploaded
Destination Container	output-data (ADF pipeline output)	✔ Created
ADF Instance	adf-week4-pipeline (V2, Central India)	✔ Created
Linked Service	LS_BlobStorage_Week4 – Azure Blob Storage	✔ Connected
Source Dataset	DS_Source_CSV – DelimitedText, header row	✔ Configured
Destination Dataset	DS_Destination_CSV – output-data container	✔ Configured
Get Metadata	Fields: Item name, Exists, Size	✔ Succeeded (6s)
Copy Data	Source → Sink with auto column mapping	✔ Succeeded (15s)
Pipeline Run	PL_Week4_CopyPipeline – Manual Trigger, 32s	✔ Succeeded
IAM Roles	Owner, Contributor, Reader, Storage Blob Data Contributor	✔ Assigned

Final Execution Screenshots

 Pipeline run: Succeeded | Data copied from input-data to output-data | Metadata validated



The screenshot displays the 'Activity runs' section for the 'PL_Week4_CopyPipeline'. The pipeline run ID is 88a25432-1ed0-49b4-b447-4480dd09a595. The activity runs table shows two activities: 'Copy data1' and 'Get Metadata1', both of which succeeded.

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User principal
Copy data1	Succeeded	Copy data	6/15/2026, 12:40:59 AM	15s	AutoResolveIntegrationRuntime (Central India)	
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 12:40:52 AM	6s	AutoResolveIntegrationRuntime (Central India)	

Summary

- Created Azure Resource Group rg-week4-pipeline under Azure for Students (\$100 free credits, no card required).
- Provisioned Storage Account week4storageadf in Central India (East US blocked for student subscription). Created input-data and output-data containers.
- Uploaded Sample - Superstore.csv (2.18 MiB) to input-data container as source file for the pipeline.
- Created ADF instance adf-week4-pipeline and explored all three tabs: Author (pipeline canvas), Monitor (run history), Manage (linked services).
- Created Linked Service LS_BlobStorage_Week4 connecting ADF to Blob Storage via Account Key authentication.
- Created source dataset DS_Source_CSV (input-data) and destination dataset DS_Destination_CSV (output-data), both DelimitedText format.
- Built pipeline PL_Week4_CopyPipeline: Get Metadata (validates Item name, Exists, Size) → Copy Data (copies file to output-data) with success dependency.
- Executed via Debug run then Trigger Now. Get Metadata: 6s, Copy Data: 15s, total pipeline: 32 seconds. Both Succeeded.
- Verified output: Sample - Superstore.csv (2.18 MiB) successfully copied to output-data container at 12:41:10 AM on 6/15/2026.
- Assigned IAM roles: Contributor and Reader on Resource Group; Storage Blob Data Contributor to ADF Managed Identity for secure, key-free access.